

HOMEWORK 2

```
library(tidyverse)
library(igraph)
library(igraphdata)
library(jsonlite)
library(visNetwork)
library(widgetframe)
library(ggthemes)
library(readr)
```

The data that I'll use for our link prediction network analysis comes from the paper "Information dynamics shape the sexual networks of Internet-mediated prostitution", from Rocha, Liljeros and Holme. I chose this network because predicting the links between sex workers and sex consumers can reinforce the findings of the paper. The paper found that there are full feedback loops when a sex worker gets positive entries, since the escorts get more sex buyers afterwards, existing attachment preference as an effect of the network.

The data was took from a web community in which female sex workers offered their services and male sex buyers wrote reviews of their encounters. As I said, the researchers found when analysing the resulting bi-partite network that a full feedback loop occurred when a sex worker gets a positive post. This means that the sex sellers who had the most positive reviews, will most likely have the most edges to them. Additionally, according to the researchers, sex buyers stay active in the forum longer than sellers. Taken both into account, it is likely that sex workers with high degree will be much more connected with high degree sex consumers. In order to see this, I'll first analyse the bi-partite network before projecting it to do the link prediction.

```
raw_data <- read_delim(
  file      = "brazilian_prostitution.csv",
  delim     = ";",
  comment   = "#",
  col_names = FALSE,
  trim_ws   = TRUE,
```

```

col_types = cols(.default = col_integer())
) |>
  rename(
    `sex_worker-id`
    `sex_buyer-id`
    `Date (in days) of posting`
    `Grade`
    `Anal sex with/without condom?`
    `Oral sex with condom?`
    `Mouth kiss?`
  )

raw_data

```

```

= X1,
= X2,
= X3,
= X4,
= X5,
= X6,
= X7)

```

```

# A tibble: 50,632 x 7
  `sex_worker-id` `sex_buyer-id` `Date (in days) of posting` Grade
      <int>         <int>         <int> <int>
1           0           1           0      0
2           2           1           0      1
3           3           1           0      0
4           0           4          21      0
5           5           6          47      0
6           5           6          73      0
7           7           8         105      0
8           9          10         121      0
9          11          12         124      1
10         13          12         126      1
# i 50,622 more rows
# i 3 more variables: `Anal sex with/without condom?` <int>,
#   `Oral sex with condom?` <int>, `Mouth kiss?` <int>

```

The paper suggests that the more time services are provided by a sex worker, the more risky sexual behaviour they permit in their encounters with clients. This might be because of pressures to either the clients themselves or to keep up with the the market, which means more exposition to sexual transmission diseases. They also found that the time they offer sexual services is relatively small, which is shown by the linear preferential attachment in short time scales and the lack of linearity in larger time frames of sex workers. Taking all this into consideration, let's start the analysis itself by taking out the multiple edges of the graph and analyse some parameters.

```
g <- graph_from_data_frame(raw_data, directed=FALSE) |> simplify()

vcount(g)
```

```
[1] 16730
```

```
ecount(g)
```

```
[1] 39044
```

```
degree(g) |>
  mean()
```

```
[1] 4.667543
```

```
assortativity_degree(g)
```

```
[1] -0.1097364
```

Our network consists of 16.730 nodes and 39.044 edges after the suppression of multiple edges. We can see that active sex buyers have a tendency to have sexual encounters with less active sex workers or viceversa. To see which set of the network drives the disassortativity, I asked the AI Claude to create a function that can measure the mean degree of the partners of each sex worker or sex buyer as a separate category.

```
# Identify each type of node
sex_workers <- as.character(unique(raw_data$`sex_worker-id`))
sex_buyers <- as.character(unique(raw_data$`sex_buyer-id`))

# For each sex buyer, calculate the average degree of the sex workers they connect to
buyer_avg <- sapply(sex_buyers, function(b) {
  partners <- neighbors(g, b)          # sex workers connected to this buyer
  mean(degree(g, partners))           # average degree of those workers
})

# For each sex worker, calculate the average degree of the sex buyers they connect to
worker_avg <- sapply(sex_workers, function(w) {
  partners <- neighbors(g, w)          # sex buyers connected to this worker
```

```
mean(degree(g, partners))          # average degree of those buyers  
})
```

```
# Compare the averages
```

```
mean(buyer_avg)    # average degree of workers that buyers connect to
```

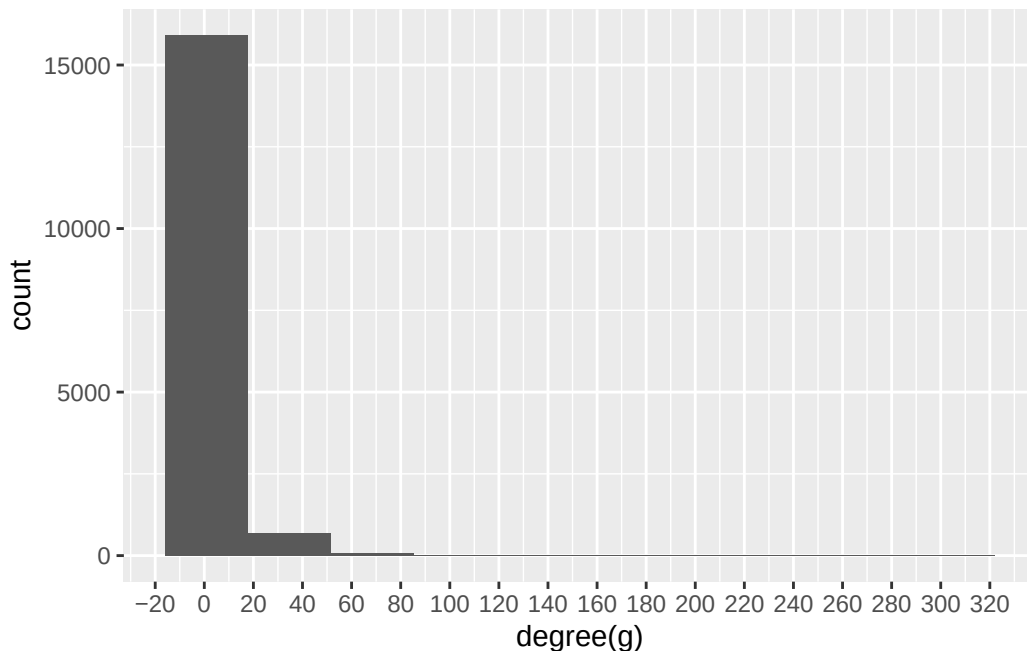
```
[1] 43.65766
```

```
mean(worker_avg)    # average degree of buyers that workers connect to
```

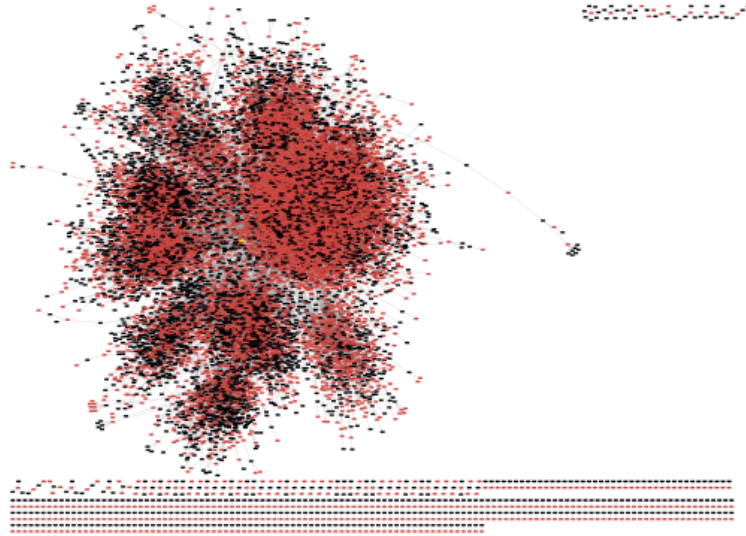
```
[1] 14.93416
```

It seems that Buyers tend to connect more with more connected sex workers and that sex workers tend to connect more with less connected buyers. This may be because sex workers have by default more connections, since they are the ones that offer their services. Also, the mean degree of each node is close to 4. Having more than 20 connections is extremely rare in this network, which further explains the disassortativity.

```
ggplot() +  
  geom_histogram(aes(x = degree(g)), bins = 10) +  
  scale_x_continuous(n.breaks = 20)
```



This is the visual representation of the bi-partite network. As we can see, there is a significant amount of isolated nodes, both in the outskirts of the giant component and in stand-alone components.



In order to perform our link prediction, we have to create a projected version of the network, since there are no triangles in a bi-partite network.

```
V(g)$type <- V(g)$name %in% as.character(unique(raw_data$`sex_worker-id`))  
  
# Project onto sex buyers (type = FALSE)  
g <- bipartite_projection(g, which = FALSE)
```

What we have now is a network where two sex buyers are connected if they have had a sexual encounter with the same sex worker. With this transformation, the main parameters have changed as well. Now there are 10,106 nodes, 668,183 edges and the mean degree is 132, very different to the prior network.

```
vcount(g)
```

```
[1] 10106
```

```
ecount(g)
```

```
[1] 668183
```

```
degree(g) |>  
  mean()
```

```
[1] 132.2349
```

Why does that change occur? This is because all edges to a given sex worker are now connected to each other, which creates a lot of connected clusters.

Now we can delete links in order to create the link prediction model.

```
set.seed(111)  
links_deleted <- sample(1:ecount(g), 5000)  
positive_class_links <- as.data.frame(ends(g, links_deleted))  
positive_class_links
```

	V1	V2
1	1932	8656
2	1049	4779
3	6182	7564
4	29	2455
5	2257	5781
6	6217	9632
7	9962	15121
8	5905	10340
9	2712	7919
10	212	4289
11	8349	9944
12	5085	5228
13	2118	2234
14	3515	10410
15	1809	12760
16	4117	14253
17	2950	9840
18	2712	9184
19	1778	8573
20	7914	11742
21	10594	10868
22	12194	14984
23	4458	11349
24	2489	3151
25	147	240

26	11383	11384
27	9205	10902
28	1861	6865
29	406	1292
30	6558	8429
31	169	8752
32	7564	11558
33	12313	12689
34	2387	6875
35	1075	4438
36	6865	8490
37	4165	14365
38	4147	4845
39	8427	13477
40	752	10446
41	1891	8247
42	5562	8364
43	4204	9501
44	4287	9551
45	10793	12174
46	1873	12517
47	5182	7422
48	12082	12212
49	4288	15653
50	551	11417
51	7039	12891
52	5699	14171
53	11568	14501
54	3460	8347
55	1370	4396
56	1056	1580
57	3292	3532
58	2649	5085
59	1553	3530
60	7963	8814
61	994	10989
62	299	5085
63	11848	14974
64	1067	6736
65	6185	11592
66	13130	14001
67	1312	7529
68	50	9205

69	2489	9629
70	4147	10230
71	1530	4032
72	467	1276
73	4432	5274
74	1303	3457
75	1952	3569
76	10659	13467
77	6558	15653
78	3628	12472
79	284	7849
80	8686	12718
81	4536	11205
82	1304	2054
83	12875	13860
84	6453	8911
85	8892	16069
86	211	1604
87	3423	5121
88	10858	14466
89	7291	8684
90	6522	14368
91	932	15459
92	156	4031
93	1313	3604
94	109	1943
95	1118	3961
96	12833	15806
97	1062	13835
98	579	1841
99	4074	6551
100	6217	15535
101	924	10595
102	4960	11095
103	271	383
104	5199	6022
105	537	4396
106	6262	9593
107	3521	3649
108	66	6950
109	145	3200
110	370	4636
111	1870	4369

112	2203	4128
113	701	11835
114	4518	12997
115	16261	16262
116	7091	13645
117	3945	8146
118	1624	10197
119	551	5658
120	4572	14843
121	649	7124
122	338	13906
123	447	9603
124	2489	16453
125	6443	11945
126	879	2531
127	10011	10169
128	2980	5830
129	2675	4394
130	14558	15943
131	12386	13448
132	419	2621
133	6105	6573
134	494	4736
135	1064	1890
136	9367	10696
137	2069	2324
138	6109	15620
139	2281	14157
140	6849	9838
141	5238	14717
142	1605	4830
143	11221	13058
144	6573	10163
145	6292	6825
146	5985	12958
147	4681	5908
148	212	10694
149	1436	5296
150	1721	4969
151	6558	8167
152	5041	8784
153	8487	16665
154	4396	12718

155	15301	16575
156	4992	15778
157	8900	9573
158	8799	9243
159	552	6182
160	169	13081
161	1815	2583
162	8445	8660
163	128	9951
164	12190	13034
165	222	631
166	8585	11877
167	141	5229
168	6556	10842
169	14439	14951
170	9635	12486
171	12174	13731
172	3467	8405
173	694	7293
174	13006	13313
175	99	574
176	3620	10757
177	7412	10008
178	8206	11348
179	4255	8177
180	5910	14881
181	1353	2858
182	4048	8627
183	50	986
184	8600	15121
185	2705	6331
186	9465	12191
187	1523	4253
188	2065	5297
189	924	4799
190	6309	7516
191	3207	5308
192	4736	5272
193	13170	13577
194	3292	6736
195	418	6558
196	1809	12800
197	6268	7800

198	7633	8347
199	924	5670
200	2301	3963
201	6701	11250
202	2034	15516
203	1952	9456
204	10869	12563
205	631	3299
206	5713	12781
207	2563	7193
208	6867	8718
209	99	13385
210	13005	13190
211	5308	9860
212	5099	7903
213	9258	14474
214	1605	2054
215	8866	8989
216	3339	5248
217	1282	4736
218	5847	6320
219	1323	1524
220	5008	13777
221	7633	8797
222	3091	13467
223	1300	11946
224	1590	3870
225	653	1656
226	1353	3672
227	714	6266
228	1641	8696
229	523	13052
230	1353	6749
231	588	915
232	2774	3575
233	3998	9465
234	6664	16340
235	3907	4105
236	4657	9838
237	831	9426
238	11055	13811
239	731	3184
240	5946	7357

241	13548	16663
242	284	12538
243	12550	12997
244	1946	4571
245	1074	7576
246	1283	2609
247	2531	13996
248	6734	7582
249	230	376
250	3756	7909
251	846	12891
252	11058	15085
253	3533	15565
254	2034	6468
255	250	1554
256	2139	15095
257	2571	6573
258	1067	13555
259	3508	4539
260	9838	13591
261	2993	4115
262	485	6726
263	3249	3903
264	30	1011
265	3340	11117
266	1646	2118
267	1891	2026
268	4240	4643
269	3384	8347
270	824	1151
271	3598	4074
272	6028	11780
273	5514	12191
274	6573	9919
275	5230	12070
276	6412	11348
277	5426	8257
278	1471	7216
279	4801	11332
280	1748	2402
281	10314	10454
282	1441	13667
283	2297	4060

284	10793	10915
285	2328	6037
286	4922	10972
287	40	1003
288	358	8421
289	6815	9450
290	1162	6816
291	7276	11780
292	2054	7887
293	8186	13837
294	2144	2266
295	2439	7106
296	4863	16410
297	154	1422
298	10066	14928
299	1211	5161
300	5085	13465
301	4287	6800
302	2568	4619
303	5909	7574
304	8989	11383
305	551	13081
306	2744	14843
307	2362	8524
308	51	4847
309	11896	14449
310	4736	11780
311	1616	1656
312	13545	15136
313	1778	11698
314	1717	7313
315	457	5600
316	171	3643
317	2938	6546
318	2819	6546
319	13280	14458
320	6468	6715
321	6171	7979
322	7831	10972
323	1803	10110
324	714	4636
325	12732	13704
326	5423	13148

327	7913	8692
328	1143	2258
329	1468	10229
330	633	13902
331	12891	14951
332	1176	2755
333	3142	5488
334	5513	6132
335	1653	5416
336	1780	1835
337	5905	6800
338	1582	7194
339	393	526
340	5974	9962
341	9725	15205
342	9689	13588
343	3507	14181
344	540	5624
345	5038	9550
346	781	8718
347	1921	7310
348	996	3311
349	239	2259
350	7569	11290
351	8026	11974
352	524	2409
353	2568	13788
354	11490	15678
355	457	6071
356	1671	8348
357	2301	12177
358	5664	10422
359	9398	11516
360	589	10055
361	10281	13509
362	3837	9675
363	5899	13697
364	11010	11946
365	10704	15504
366	1653	7979
367	2587	4187
368	812	6558
369	4855	7582

370	1089	9987
371	86	4387
372	346	7636
373	1059	2375
374	1	4945
375	4524	13704
376	2678	10697
377	3508	7124
378	485	610
379	1153	2708
380	454	2719
381	1078	13899
382	467	6278
383	309	12757
384	2581	4294
385	9425	12190
386	10895	13477
387	11946	15504
388	4289	11594
389	1956	7168
390	1392	6503
391	211	9395
392	1144	1175
393	7261	10859
394	472	1709
395	5069	14460
396	4518	9136
397	589	8247
398	3578	5985
399	274	1814
400	4353	16189
401	1076	3340
402	3292	5388
403	2755	8680
404	7892	8464
405	4868	5308
406	10609	12440
407	10352	14566
408	2411	13447
409	1436	2368
410	540	1803
411	5069	8085
412	8678	9879

413	212	228
414	1696	8923
415	6573	9931
416	3575	4115
417	86	11336
418	1293	11392
419	2609	8923
420	7979	14640
421	13635	15653
422	551	9570
423	108	10869
424	358	9838
425	4259	13031
426	406	9465
427	6572	10065
428	3517	5450
429	3073	4165
430	2606	6183
431	270	14495
432	10453	12388
433	4147	4173
434	3260	8962
435	52	58
436	6443	6578
437	8130	12465
438	7055	13897
439	7215	9619
440	3881	6950
441	4218	11698
442	10721	13545
443	7583	15564
444	3828	6147
445	1579	3151
446	2719	8551
447	1430	7215
448	3290	11783
449	11057	14470
450	760	14157
451	3162	10546
452	8630	11990
453	1768	10779
454	10726	13646
455	3786	4664

456	1313	7603
457	2144	12687
458	10268	13405
459	11216	11946
460	1824	6035
461	3518	7337
462	846	1997
463	9693	14336
464	145	6537
465	6522	10869
466	58	1109
467	54	11187
468	2265	6010
469	2455	4514
470	955	4282
471	5121	15682
472	2531	4645
473	3753	10078
474	6537	12074
475	1297	2639
476	2609	9649
477	10805	14696
478	3312	9328
479	2639	10815
480	1364	8827
481	1521	11600
482	1193	4255
483	1052	11144
484	2245	3682
485	1431	3611
486	3074	10858
487	5412	8219
488	1744	5599
489	7488	12377
490	3020	13537
491	8652	11043
492	7902	13448
493	4467	15851
494	4846	6022
495	1891	3479
496	1960	13645
497	321	5713
498	13477	14780

499	2063	2665
500	8678	9792
501	4892	9974
502	701	2164
503	15461	16474
504	8690	13122
505	7311	12423
506	1858	15647
507	2531	3174
508	831	1720
509	2048	8425
510	4117	14156
511	1932	6126
512	7332	9308
513	3207	14495
514	1144	2983
515	1898	12745
516	3962	8757
517	2938	6344
518	129	4873
519	1676	5069
520	3292	3672
521	5006	7918
522	2649	9792
523	7250	9053
524	7842	16037
525	2368	2560
526	3426	5342
527	4202	9425
528	2425	5384
529	5713	5959
530	7509	12401
531	2469	5817
532	5908	13509
533	8425	14425
534	8924	14698
535	1841	2990
536	4861	14223
537	3518	5121
538	3794	4482
539	5311	8430
540	7250	11493
541	7213	10789

542	6624	10945
543	1812	7595
544	6558	9200
545	6268	10145
546	6167	15851
547	9973	14919
548	8656	10419
549	6071	10790
550	305	1334
551	3485	6737
552	2063	10057
553	734	3299
554	3761	14843
555	4402	4603
556	2314	5548
557	10394	14368
558	1662	1835
559	4204	7620
560	10213	11402
561	156	1161
562	744	9359
563	13478	15424
564	1205	9982
565	8155	13034
566	1656	8257
567	3373	13413
568	1562	11793
569	5685	5908
570	5327	7919
571	687	3942
572	5258	9725
573	4737	9657
574	657	6642
575	5161	11640
576	4646	8347
577	2705	5715
578	1274	1970
579	4481	10476
580	1552	2906
581	1075	1258
582	5513	8614
583	1422	5919
584	270	13281

585	1454	8405
586	8182	10174
587	3231	3322
588	907	5086
589	2712	4294
590	13190	15026
591	4643	5098
592	350	5346
593	1809	4605
594	33	5121
595	11135	15254
596	2923	11348
597	1551	3539
598	4153	5199
599	1118	1669
600	895	986
601	2387	8316
602	212	6126
603	1056	9475
604	156	2368
605	8692	11216
606	2194	9857
607	5423	10057
608	8058	9103
609	4032	11053
610	5269	11205
611	1649	3063
612	4692	9243
613	9617	11051
614	1323	10827
615	5085	12989
616	12779	12933
617	781	11590
618	402	3532
619	11641	12691
620	4683	10066
621	5226	9033
622	1861	10198
623	7422	9221
624	3929	4641
625	3473	4412
626	6218	12010
627	7124	11095

628	1840	4742
629	4766	5809
630	3089	9738
631	937	5817
632	354	4084
633	1441	3940
634	7273	15121
635	331	548
636	6385	8101
637	7529	13036
638	338	1202
639	7757	14099
640	2531	6439
641	3598	7484
642	8039	8850
643	7582	7608
644	701	2712
645	913	7672
646	7150	14263
647	9395	12314
648	5652	5975
649	1523	1684
650	1817	3926
651	136	8347
652	5442	12901
653	7216	8386
654	8828	8959
655	2459	8610
656	9649	13545
657	7276	11539
658	10149	10488
659	10446	16533
660	3460	8246
661	4147	7311
662	1312	12440
663	3299	8989
664	2589	8140
665	6701	15152
666	10972	14640
667	8450	12788
668	3257	11306
669	2430	12874
670	5905	7529

671	5069	12898
672	471	5809
673	9551	14952
674	4255	14137
675	6459	9977
676	3520	11026
677	3963	11945
678	4970	13132
679	9792	11794
680	363	2235
681	6558	16344
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1542	7216	13887
1543	732	1377
1544	7484	8965
1545	1363	4106
1546	1921	12294
1547	5101	9114
1548	2758	8678
1549	5549	10185
1550	4740	8708
1551	523	1767
1552	6350	13929
1553	162	7215
1554	4726	16546
1555	147	1697
1556	1	370
1557	12589	12988
1558	4485	5977
1559	2088	15712
1560	3123	8799
1561	2162	14843
1562	7340	13467
1563	3485	15535
1564	4218	12834
1565	604	12402
1566	918	9262
1567	8047	15522
1568	4858	10828
1569	3539	9580
1570	8731	8798
1571	5652	10600
1572	5624	7150
1573	1605	5548

1574	14210	14717
1575	499	6321
1576	11524	13809
1577	3963	6060
1578	7509	12875
1579	958	5311
1580	1312	6248
1581	1778	16509
1582	2965	11922
1583	516	1441
1584	5279	6781
1585	9580	13170
1586	470	10931
1587	5239	6989
1588	29	7359
1589	6751	8221
1590	2789	4964
1591	2151	8491
1592	8698	13903
1593	4965	8147
1594	1356	11095
1595	11493	12010
1596	9395	13148
1597	6578	7325
1598	1149	9877
1599	1590	6734
1600	2404	6027
1601	1861	4797
1602	1940	13977
1603	932	1441
1604	9350	9359
1605	1490	4635
1606	5094	6459
1607	929	8881
1608	8487	12803
1609	6459	14772
1610	475	4606
1611	3652	9912
1612	3918	5121
1613	9105	14008
1614	6051	6362
1615	1355	11304
1616	1580	15238

1617	5388	11898
1618	8488	8846
1619	166	5835
1620	124	7359
1621	1684	6848
1622	4294	6692
1623	7979	8757
1624	4636	8316
1625	6800	14474
1626	4511	9082
1627	1007	3690
1628	4217	10479
1629	8678	11890
1630	3123	4397
1631	5395	12589
1632	86	649
1633	4172	13835
1634	3204	9129
1635	10189	11997
1636	2035	9465
1637	1049	1737
1638	2609	4774
1639	1427	13434
1640	1997	2165
1641	13116	15122
1642	12824	16282
1643	1742	5847
1644	4032	10229
1645	1195	5997
1646	19	13558
1647	2980	13148
1648	7148	11783
1649	6373	14457
1650	86	13448
1651	471	1946
1652	1468	1803
1653	5101	7224
1654	4858	13265
1655	6017	15825
1656	1580	14524
1657	489	11896
1658	10453	16100
1659	3339	13562

1660	2469	8646
1661	5083	14263
1662	3211	15248
1663	2169	3918
1664	1504	9700
1665	1684	7959
1666	1089	2706
1667	9444	14403
1668	1278	5513
1669	156	14904
1670	8061	8989
1671	8186	13573
1672	1310	3406
1673	1287	7525
1674	6268	11010
1675	395	7055
1676	2564	6648
1677	4570	15969
1678	1385	7216
1679	2746	10221
1680	6868	14928
1681	8488	13788
1682	2975	4829
1683	8316	13888
1684	378	15535
1685	2118	12203
1686	2554	2937
1687	601	788
1688	5701	10917
1689	6418	8147
1690	9022	10007
1691	1303	2744
1692	122	2595
1693	2444	5913
1694	4389	11199
1695	1669	8257
1696	6392	8678
1697	2297	14879
1698	12283	12545
1699	10034	12682
1700	192	547
1701	1876	6950
1702	13740	14179

1703	2654	15121
1704	5327	14018
1705	1653	16618
1706	5083	11949
1707	2927	11945
1708	7303	9167
1709	9396	10448
1710	9586	13128
1711	2344	6207
1712	5101	11493
1713	846	15620
1714	3961	15565
1715	3520	4824
1716	19	11945
1717	6383	10670
1718	2469	16280
1719	5567	13838
1720	5807	8382
1721	4812	11057
1722	2712	5794
1723	6527	7303
1724	212	3207
1725	1106	3998
1726	2141	3974
1727	4295	10881
1728	3930	5653
1729	6802	10065
1730	2151	7196
1731	8520	14395
1732	6916	10490
1733	371	4113
1734	1385	4605
1735	13256	13359
1736	1065	1293
1737	308	12379
1738	12907	13465
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1740	7488	12010
1741	2297	6454
1742	11221	14646
1743	2430	2709
1744	4607	5188
1745	5685	13821

1746	196	2927
1747	154	5182
1748	1297	7039
1749	3216	11338
1750	471	858
1751	4537	14688
1752	1732	1946
1753	4521	5713
1754	7580	9361
1755	2002	4772
1756	6324	8518
1757	1780	4400
1758	3182	4031
1759	3460	5051
1760	257	4771
1761	1601	13984
1762	1356	7091
1763	3686	13589
1764	3396	10065
1765	377	9838
1766	1300	1875
1767	1653	2998
1768	2259	6060
1769	1452	1502
1770	58	154
1771	3142	5186
1772	6594	12945
1773	914	9604
1774	2162	8791
1775	505	12439
1776	694	15778
1777	12	10065
1778	1203	13081
1779	5913	12478
1780	184	3168
1781	4059	14540
1782	8526	9838
1783	3396	5425
1784	932	3633
1785	5083	15026
1786	1113	1493
1787	5388	5416
1788	1064	10020

1789	3217	7835
1790	1278	3672
1791	10925	16340
1792	3173	3578
1793	11808	13980
1794	43	213
1795	7569	11402
1796	5735	11133
1797	5098	7359
1798	7893	11662
1799	1001	1436
1800	5458	15099
1801	490	7580
1802	1056	1064
1803	7580	12427
1804	3650	5314
1805	12450	16374
1806	9088	13280
1807	1011	6726
1808	2861	14274
1809	824	2221
1810	8550	8791
1811	298	5052
1812	419	1684
1813	5567	8963
1814	279	2952
1815	13316	15535
1816	4218	12891
1817	2118	10281
1818	538	4089
1819	5228	5850
1820	7001	10277
1821	5274	8286
1822	5083	14020
1823	12600	14299
1824	3794	13301
1825	2655	6551
1826	1062	1955
1827	1078	13915
1828	66	1547
1829	4460	10545
1830	4537	6467
1831	66	2258

1832	862	13778
1833	3508	4505
1834	11783	13465
1835	9999	11057
1836	395	5055
1837	407	1065
1838	1356	13555
1839	8040	8469
1840	966	1907
1841	8892	15777
1842	6309	11947
1843	6016	10678
1844	7340	11526
1845	9446	14299
1846	4258	8680
1847	66	1106
1848	4294	5311
1849	2751	10374
1850	6648	13039
1851	8866	9116
1852	1120	4760
1853	551	5822
1854	781	1551
1855	4202	7432
1856	566	13915
1857	350	6558
1858	4141	8840
1859	5414	12729
1860	1579	5548
1861	2947	3945
1862	10277	13264
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1866	913	1095
1867	8523	10374
1868	4951	7905
1869	165	1062
1870	4282	4797
1871	10784	13191
1872	4973	10221
1873	4173	5977
1874	649	16037

1875	1118	4554
1876	2368	6610
1877	2775	6429
1878	1144	3532
1879	1605	10278
1880	489	5117
1881	2192	10664
1882	3892	11736
1883	2177	3791
1884	5342	15520
1885	3963	11216
1886	6148	6627
1887	1078	9858
1888	924	3534
1889	13300	14551
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1891	1067	16106
1892	10724	12402
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1894	9464	11338
1895	5138	9773
1896	12440	16216
1897	352	1297
1898	4243	8990
1899	8464	8492
1900	5028	13047
1901	5517	7308
1902	11698	14001
1903	494	3661
1904	15375	16314
1905	8221	8421
1906	2710	8930
1907	1144	8566
1908	6783	10826
1909	5029	10631
1910	4217	6783
1911	2413	3195
1912	6439	12010
1913	1979	11303
1914	8964	15876
1915	3207	4799
1916	12732	14952
1917	1997	4646

1918	3071	10164
1919	4975	13297
1920	7757	11043
1921	1816	7614
1922	5238	8489
1923	1283	4170
1924	6385	14671
1925	931	12511
1926	559	3204
1927	2609	2891
1928	2455	8498
1929	3413	5342
1930	9879	10600
1931	494	10813
1932	5385	8233
1933	12258	15762
1934	6573	9573
1935	4112	9573
1936	1441	9525
1937	4946	6817
1938	2343	10973
1939	4570	6868
1940	8430	12801
1941	10207	12841
1942	7357	9022
1943	7593	8141
1944	567	3787
1945	932	13477
1946	2425	12855
1947	304	1490
1948	1080	9573
1949	11629	13081
1950	9671	10586
1951	4742	6022
1952	2786	7248
1953	6385	7175
1954	6429	9966
1955	1282	12251
1956	6868	9843
1957	931	4760
1958	4646	6952
1959	10064	12978
1960	3340	4479

1961	277	1468
1962	4485	6375
1963	2010	16476
1964	2407	15458
1965	2967	4147
1966	13537	15907
1967	156	2635
1968	13180	15296
1969	996	1312
1970	4048	6349
1971	175	2425
1972	1078	14039
1973	996	14065
1974	349	16100
1975	1067	8316
1976	1774	11826
1977	3020	3432
1978	1065	7886
1979	3759	6880
1980	1203	12690
1981	13465	14540
1982	91	6344
1983	6504	9936
1984	5713	8618
1985	1605	8061
1986	4726	8086
1987	2675	11150
1988	2143	6558
1989	10054	13410
1990	1698	5549
1991	2068	14780
1992	13897	14261
1993	3174	9725
1994	1835	14902
1995	3274	7672
1996	5416	10066
1997	7303	10327
1998	967	2953
1999	1760	1962
2000	8608	12827
2001	3216	12650
2002	5830	10065
2003	4853	6351

2004	8925	11292
2005	1011	5297
2006	13549	14348
2007	2267	15226
2008	12093	13384
2009	9424	9773
2010	640	1468
2011	4523	4660
2012	5691	7215
2013	2469	4612
2014	109	924
2015	239	1760
2016	5086	7517
2017	508	4369
2018	2508	5875
2019	5038	10112
2020	5055	5701
2021	8512	15289
2022	7003	13048
2023	13537	14672
2024	4289	11698
2025	902	986
2026	3783	9243
2027	1377	10223
2028	9078	12487
2029	8302	9022
2030	11264	11366
2031	2118	12379
2032	1149	15851
2033	395	11057
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2037	395	3101
2038	4295	6817
2039	1522	2992
2040	7091	11725
2041	5485	6751
2042	3186	9200
2043	966	3518
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2045	5513	8446
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2047	1297	9303
2048	1641	4766
2049	1991	11166
2050	6624	14395
2051	958	1653
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2061	1323	3439
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2064	175	8614
2065	5860	7580
2066	2343	8990
2067	1818	1946
2068	196	1769
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2070	2325	2717
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2072	12800	13409
2073	5157	5863
2074	1468	4919
2075	2304	8737
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2077	91	6217
2078	9456	15373
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2080	8298	8425
2081	5809	14018
2082	8738	12937
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2084	6817	8627
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2092	7055	7886
2093	2438	6671
2094	2469	4061
2095	5197	15119
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2100	7484	12192
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2102	6824	15133
2103	1840	9446
2104	352	8869
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2106	5052	6936
2107	2301	11290
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2109	7276	10066
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2111	902	1979
2112	10956	15296
2113	3285	9657
2114	7458	13611
2115	3688	15108
2116	2583	9858
2117	1991	4154
2118	4620	10820
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2127	1645	5327
2128	2301	4969
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2130	2072	2193
2131	2469	14222
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2136	467	4147
2137	10973	11306
2138	12652	14686
2139	69	5670
2140	3059	7027
2141	1441	6633
2142	358	2708
2143	7757	13898
2144	2583	13370
2145	13996	14095
2146	5975	14235
2147	6157	14951
2148	2819	11338
2149	2705	4374
2150	1157	4992
2151	193	3787
2152	1153	6725
2153	2040	9940
2154	15252	16097
2155	4007	10266
2156	865	13537
2157	7340	12683
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2159	2068	10895
2160	11274	13491
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2162	9056	9422
2163	2245	6268
2164	9267	16672
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2172	3908	6217
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2178	742	7105
2179	4255	10743
2180	2151	6825
2181	1377	8066
2182	3413	6558
2183	2649	3591
2184	3900	4170
2185	1543	13389
2186	12250	14524
2187	33	376
2188	1080	4239
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2190	3654	4922
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2192	3416	6472
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2199	211	1313
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2206	8599	12174
2207	1422	6468
2208	3174	11933
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2211	11454	13314
2212	6556	7485
2213	345	14046
2214	937	3291
2215	5363	11173
2216	12481	13116
2217	3732	4518
2218	182	6547

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2220	1818	10785
2221	8708	13903
2222	1049	4258
2223	443	2222
2224	2689	6330
2225	10868	11287
2226	3056	8040
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2228	5947	8967
2229	10972	13526
2230	585	4266
2231	3349	10202
2232	211	16340
2233	6294	8768
2234	11725	13829
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2236	9068	12406
2237	3769	4139
2238	393	15955
2239	5083	6330
2240	6454	15825
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2242	752	6159
2243	4845	4946
2244	2344	7488
2245	10221	16037
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2247	3633	15052
2248	7248	14253
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2251	8026	11080
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2254	2425	6840
2255	58	924
2256	781	11095
2257	309	1353
2258	3916	5272
2259	14464	15762
2260	1076	2343
2261	516	1589

2262	1471	4579
2263	5699	8678
2264	3211	10281
2265	2101	2123
2266	9114	10036
2267	2281	7484
2268	2455	3768
2269	284	9910
2270	1203	4868
2271	781	918
2272	1056	1454
2273	1955	10211
2274	2644	6950
2275	1064	2786
2276	7580	12192
2277	3940	7055
2278	467	10363
2279	973	6016
2280	742	5482
2281	10057	10568
2282	7792	14524
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2285	5069	5912
2286	6825	10898
2287	5691	14246
2288	2967	13046
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2291	9339	14879
2292	137	4060
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2296	3544	6802
2297	2180	6518
2298	3578	11953
2299	1656	15132
2300	1890	7488
2301	5937	13500
2302	1451	3143
2303	2729	5121
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2305	8350	10691
2306	4608	9580
2307	2047	15180
2308	8366	14766
2309	508	3905
2310	1314	5848
2311	294	1803
2312	2035	2729
2313	1078	9393
2314	1054	3252
2315	11132	13583
2316	2982	10278
2317	2234	3271
2318	3657	5021
2319	494	1952
2320	6472	13537
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2322	619	5072
2323	6934	7740
2324	5671	7039
2325	4682	7251
2326	5733	6558
2327	1656	5185
2328	3507	5269
2329	2343	15890
2330	1011	1890
2331	6715	10915
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2335	4461	7613
2336	211	8146
2337	175	3533
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2339	846	2861
2340	3299	5407
2341	7173	12408
2342	3522	4863
2343	2854	7140
2344	3073	4735
2345	2705	10171
2346	8576	8814
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2348	1062	13896
2349	2850	5988
2350	1960	9629
2351	3222	6132
2352	1653	5227
2353	5423	11449
2354	588	11470
2355	1815	5041
2356	393	9584
2357	2453	4886
2358	4173	10473
2359	10221	16279
2360	5368	12271
2361	1075	2151
2362	5908	9475
2363	2719	5459
2364	8614	11338
2365	5038	6884
2366	2705	11053
2367	4049	13081
2368	108	3596
2369	386	2797
2370	12879	13637
2371	4966	12465
2372	90	3906
2373	7607	7796
2374	4202	8064
2375	161	8967
2376	2531	12719
2377	2705	3467
2378	1579	1945
2379	9027	10724
2380	11861	13204
2381	683	9022
2382	3508	6875
2383	32	50
2384	238	887
2385	99	5407
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2387	4481	16344
2388	358	2814
2389	525	4458
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2396	2086	3499
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2399	2368	7511
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2414	2538	5797
2415	1063	2774
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2420	11889	14566
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2427	7213	10211
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2429	2368	5067
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2439	2245	13383
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2461	33	2151
2462	3498	6395
2463	9525	9636
2464	928	9617
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2473	2758	6664
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3224	407	13052
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3226	6373	12404
3227	3485	9032
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3230	15226	16100
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3236	2923	3633
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3239	2035	9328
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3241	992	1778
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3243	378	9792
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3246	2274	5457
3247	6386	8102
3248	2583	11266
3249	6526	6769
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3252	5238	6886
3253	4619	13611
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3255	1812	3508
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3257	3515	8379
3258	5038	13980
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3260	1977	5197
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3263	6751	10972
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3265	10724	15778
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3267	8302	16637
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4066	376	2982
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4105	8322	9902
4106	4355	4644
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4129	1962	2056
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4135	1598	1626
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4137	3375	8379
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4142	9941	15395
4143	90	9185
4144	432	9078
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4182	1946	6615
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4217	8640	11002
4218	40	7349
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4220	2368	4937
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4223	171	6996
4224	865	15620
4225	6412	12377
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4235	2595	4218
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4244	1487	7563
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4290	2556	4864
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4374	2690	9402
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4889	69	2708
4890	910	2411
4891	829	11041
4892	14808	15876
4893	7326	7606
4894	1200	6825
4895	90	8555
4896	7513	15211
4897	1441	3613
4898	4324	7459
4899	12797	13573
4900	9465	14451
4901	1645	6817
4902	2583	4619
4903	6099	7550
4904	9792	10207
4905	3065	13558
4906	4294	11290
4907	1564	10215
4908	370	478
4909	5038	6984
4910	4760	6766
4911	1078	4560
4912	4965	8303
4913	786	3217
4914	8733	13409
4915	1415	8568
4916	4410	5350
4917	13986	14470
4918	1303	7529
4919	5101	13556
4920	4560	7513
4921	2362	15565
4922	607	5514
4923	308	349
4924	924	6324
4925	6825	8085
4926	14732	14922
4927	318	9114

4928	8990	10545
4929	4660	10584
4930	1979	10882
4931	7304	8247
4932	11946	14441
4933	3717	3816
4934	7509	16479
4935	162	3204
4936	3048	4368
4937	4798	11594
4938	1562	8718
4939	418	3151
4940	9219	9234
4941	11338	15151
4942	6735	8247
4943	737	2064
4944	1579	14157
4945	4990	10779
4946	11867	13081
4947	6865	11292
4948	4289	11235
4949	551	589
4950	351	6612
4951	734	2170
4952	3596	4363
4953	1813	5187
4954	4539	8469
4955	2022	7582
4956	5548	6029
4957	1465	2921
4958	2489	8405
4959	3114	13103
4960	5922	7215
4961	2649	7913
4962	4343	8995
4963	57	91
4964	4618	7447
4965	16	2139
4966	7530	9553
4967	5207	7308
4968	537	4960
4969	1725	2063
4970	6217	12588

4971	161	7467
4972	3299	15891
4973	8989	10631
4974	2583	6429
4975	196	2301
4976	5199	15113
4977	8125	12747
4978	5584	6217
4979	1293	4239
4980	572	6016
4981	6668	14900
4982	2722	9328
4983	750	8234
4984	3138	3794
4985	3647	5343
4986	2758	9550
4987	8418	13778
4988	15454	16209
4989	731	6767
4990	3216	4087
4991	8841	15727
4992	1193	7130
4993	4294	7791
4994	7150	8767
4995	7484	10904
4996	4863	15640
4997	5513	7749
4998	6099	10724
4999	589	5403
5000	1955	12007

Positive class links are the ones that were actually in the network, hence positive. Now we'll create an empty dataframe that will be filled with the links that we'll predict out, hence, negative class links.

```
negative_class_links <- data.frame()
```

The following is the network without the links. With this, we have two networks, `g` (with the links) and `g_prime` (without the links that we want to predict). How well the model is at predicting will depend on how close `g_prime` gets to `g`.

```
g_prime <- delete_edges(g, links_deleted)
```

The new fake links will be created by sampling edges from high degree nodes. Given that our network is really dense after the transformation and that the mean degree is 132, we need to put a number that is higher than that. The threshold of the third quantile of the distribution is 149, therefore, I'm going to settle to 150 as the threshold for the most connected nodes.

```
quantile(degree(g), 0.75)
```

```
75%  
149
```

```
most_connected <- V(g)$name[degree(g) > 150]
```

The negative class dataframe will be filled now with sampled connections between the most connected nodes. The code will create 5000 links between the most connected links.

```
negative_class_links <- data.frame(V1 = integer(), V2 = integer())  
  
while(nrow(negative_class_links) < 5000) {  
  pair <- sample(most_connected, 2)  
  
  if(are_adjacent(g, pair[1], pair[2]) == FALSE) {  
    negative_class_links <- rbind(negative_class_links,  
                                  data.frame(V1 = pair[1],  
                                              V2 = pair[2]))  
  }  
}
```

Let's store the edges into two separate dataframes and then join them with a new column which tells us if the edge actually exists or was sampled.

```
true_edges <- data.frame(positive_class_links, obs = 1)  
false_edges <- data.frame(negative_class_links, obs = 0)  
  
total_edges <- rbind(true_edges, false_edges)  
colnames(total_edges) <- c("id1", "id2", "obs")
```

Now that we have both the fake links and the existing links, we can start the process to make the link prediction model. What we need now are the features that will make it possible that the model predicts whether a link is real ($\text{obs} = 1$) or is fake ($\text{obs} = 0$). Those features will be similarity metrics between every two nodes. If the nodes that have had sexual encounters with a sex worker have more in common than the nodes that were connected with the sampling method, they will have metrics that the model can use to predict whether an edge is real or not. Otherwise, if the model doesn't differentiate when an edge is real or fake, there must be another factor outside of the similarity metrics that explains having had a sexual encounter with the same sex worker.

The similarity metrics that will be calculated are the jaccard coefficient, which shows us if two nodes have higher shared neighbors compared to the total nodes, the Adamic-Adar coefficient, which tells us if two nodes have shared sexual encounters with sex workers of less degree, and the preferential attachment coefficient, which multiplies the degree of node 1 by the degree of node 2 and tells us if well connected nodes have links with other well connected nodes.

Before calculating the similarity heuristics, let's precalculate the neighborhood of each node of the `g_prime` graph.

```
total_edges$id1 <- as.character(total_edges$id1)
total_edges$id2 <- as.character(total_edges$id2)

n1 <- neighborhood(g_prime, order=1, nodes=total_edges$id1)
n2 <- neighborhood(g_prime, order=1, nodes=total_edges$id2)
```

Now, the heuristics for each pair of nodes.

```
# Precompute degree for all nodes
all_degrees <- degree(g_prime)

# Initialize columns
total_edges$sim_jacc <- 0
total_edges$sim_aa <- 0
total_edges$sim_pref <- 0

for (i in 1:nrow(total_edges)) {

  common <- intersection(n1[[i]], n2[[i]])

  # Jaccard
  total_edges$sim_jacc[i] <- length(common) /
    length(union(n1[[i]], n2[[i]]))
```

```

# Adamic-Adar - use precomputed degrees
if (length(common) > 0) {
  deg_common <- all_degrees[as.character(common)]
  deg_common <- deg_common[deg_common > 1] # avoid log(1) = 0 -> Inf
  if (length(deg_common) > 0) {
    total_edges$sim_aa[i] <- sum(1 / log(deg_common))
  }
}

# Preferential attachment
total_edges$sim_pref[i] <- length(n1[[i]]) * length(n2[[i]])
}

```

Now that we have all our features computed for each pair of nodes, we need to split the data into a training set and a testing set. The model will learn from the training data and we'll evaluate its predictions on the test data, simulating how it would perform on truly unseen links. We use an 80/20 split — 80% of the pairs for training, 20% for testing — and fix the random seed for reproducibility. We also clean up any Inf values that may have appeared in the Adamic-Adar column from cases where common neighbors had degree 1, since logistic regression cannot handle infinite values as features. Then we will be able to train the model and perform the prediction with it. After the prediction.

```

set.seed(123)
train_index <- sample(1:nrow(total_edges), 0.8 * nrow(total_edges))

total_edges_train <- total_edges[train_index, ]
total_edges_test  <- total_edges[-train_index, ]

total_edges_train$sim_aa[is.infinite(total_edges_train$sim_aa)] <- 0
total_edges_test$sim_aa[is.infinite(total_edges_test$sim_aa)]   <- 0

total_edges_test$sim_aa[is.na(total_edges_test$sim_aa)] <- 0
total_edges_train$sim_aa[is.na(total_edges_train$sim_aa)] <- 0

# Refit the model with clean training data
glm_link <- glm(obs ~ sim_jacc + sim_aa + sim_pref,
  data      = total_edges_train,
  family    = binomial(link = 'logit'))

# Re-predict

```

```
glm_prediction <- predict.glm(glm_link,
                             newdata = total_edges_test,
                             type     = "response")
```

```
library(caret)
```

Cargando paquete requerido: lattice

Adjuntando el paquete: 'caret'

The following object is masked from 'package:purrr':

lift

```
pred <- sign(glm_prediction > 0.3)
```

```
confusionMatrix(factor(pred),factor(total_edges_test$obs),positive = "1")
```

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	783	110
1	238	869

Accuracy : 0.826
 95% CI : (0.8087, 0.8424)
 No Information Rate : 0.5105
 P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.6528

Mcnemar's Test P-Value : 9.902e-12

Sensitivity : 0.8876
 Specificity : 0.7669
 Pos Pred Value : 0.7850
 Neg Pred Value : 0.8768

```
Prevalence : 0.4895
Detection Rate : 0.4345
Detection Prevalence : 0.5535
Balanced Accuracy : 0.8273
```

```
'Positive' Class : 1
```

The model performs well overall — with an accuracy of 82.6% and a balanced accuracy of 82.7%, the three similarity heuristics combined are clearly capturing meaningful structural information about who is likely to be connected. The model is slightly better at identifying real links (sensitivity = 0.89) than at correctly rejecting non-links (specificity = 0.77), meaning it has a mild tendency to over-predict connections. With a Kappa of 0.65 and a p-value below $2.2e-16$ we can confidently say the model significantly outperforms random guessing or always predicting the majority class.

To see which coefficient is driving the model we need to look at the model coefficients. The larger the absolute value of a coefficient (and the more statistically significant), the stronger that feature's contribution to the prediction

```
summary(glm_link)
```

Call:

```
glm(formula = obs ~ sim_jacc + sim_aa + sim_pref, family = binomial(link = "logit"),
     data = total_edges_train)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.499e+00	8.094e-02	-43.231	< 2e-16 ***
sim_jacc	3.278e+01	8.575e-01	38.230	< 2e-16 ***
sim_aa	3.540e-01	3.132e-01	1.130	0.258
sim_pref	-1.249e-06	1.974e-07	-6.328	2.49e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 11090.1 on 7999 degrees of freedom
Residual deviance: 4964.6 on 7996 degrees of freedom
AIC: 4972.6
```

Number of Fisher Scoring iterations: 7

Looking at the heuristics themselves, Jaccard similarity is by far the most important predictor. This means that the proportion of shared neighbors between two buyers is the single strongest signal of a real link. Preferential attachment is also statistically significant but with smaller and negative coefficient. According to the authors of the network, “the preferential attachment for sex sellers was close to linear ($\alpha = 1$) at shorter time scales, but sublinear for longer sampling windows” For them, this reflects that sellers quit being escorts quickly. Instead, once Jaccard is accounted for, sharing many overlapping neighbors matters more than raw popularity”. This aligns with the negative assortativity we found earlier in the bi-partite network: high-degree buyers tend to share workers with lower-degree buyers, not with the other top-tier buyers. Adamic-Adar is not statistically significant in this model. Once Jaccard is included as a feature, Adamic-Adar adds little additional explanatory power, since both metrics rely on common neighbors and end up being highly correlated.

Given the findings of the authors of the paper, there are still driving forces that I pointed out earlier that could account for an even better model. For one, time is not accounted for in the model, specifically, how long the sex workers have been working as such. Given that the longer they work the riskier sex behaviour they accept and that a feedback loop is in place when the grades are high, taking both of these dimensions as attributes in the network would make it even better. Both of these dimensions have a huge impact in the spreading of sexual transmission infections and diseases, which will be the topic of my next homework assignment.